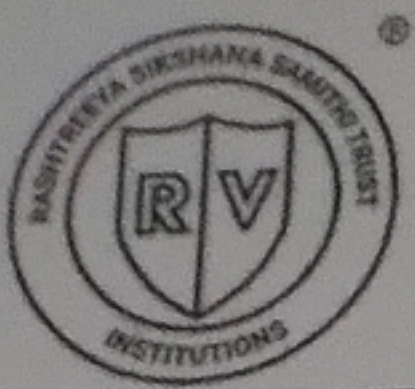




DEPARTMENT OF MATHEMATICS

Academic year 2024-2025 (Odd Sem)

(IMPROVEMENT CIE - B. E. PROGRAMS, EC, EE, EI, ET)

3	Obtain the complex Fourier series for the function $f(x) = \begin{cases} -a, & -2 < x < 0 \\ a, & 0 < x < 2 \end{cases}$ where a is a real constant, over the interval $(-2, 2)$. Deduce the corresponding trigonometric Fourier series.	10	L3	2																
4a	Find the half-range Fourier cosine series of the function $f(x) = (x + 1)^2$ in $(0, 1)$. Hence find the value of the series $1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots$.	06	L2	3																
4b	Determine the half-range Fourier sine series for the function $f(x) = e^{cx}$ for $0 < x < \pi$, where ' c ' is constant.	04	L1	3																
5	The displacement y of a part of a mechanism is tabulated with corresponding angular movement x of the function: <table><tr><td>x</td><td>0</td><td>$\frac{\pi}{3}$</td><td>$\frac{2\pi}{3}$</td><td>π</td><td>$\frac{4\pi}{3}$</td><td>$\frac{5\pi}{3}$</td><td>2π</td></tr><tr><td>y</td><td>1.0</td><td>1.4</td><td>1.9</td><td>1.7</td><td>1.5</td><td>1.2</td><td>1.0</td></tr></table> Expand y in the form of a Fourier series up to <u>second harmonic</u>. Also calculate y when $x = \frac{\pi}{4}$.	x	0	$\frac{\pi}{3}$	$\frac{2\pi}{3}$	π	$\frac{4\pi}{3}$	$\frac{5\pi}{3}$	2π	y	1.0	1.4	1.9	1.7	1.5	1.2	1.0	10	L2	
x	0	$\frac{\pi}{3}$	$\frac{2\pi}{3}$	π	$\frac{4\pi}{3}$	$\frac{5\pi}{3}$	2π													
y	1.0	1.4	1.9	1.7	1.5	1.2	1.0													

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Quiz	Max Marks	4	6	-	-	7	2	1	-	-	-
	Test		-	24	16	10	4	26	20	-	-	-



DEPARTMENT OF MATHEMATICS

Academic year 2024-2025 (Odd Sem)

(IMPROVEMENT CIE - B. E. PROGRAMS, EC, EE, EI, ET)

Date	7 th January 2024	Maximum Marks	10 +50
Time	09:30 AM to 11 :30 AM	Duration	120 Mins
Semester	III Sem B. E. (EC, EE, EI, ET)		
MA231TA – LINEAR ALGEBRA, FOURIER TRANSFORMS AND STATISTICS			

Q. No.	PART - A	M	BT	CO
1	Fourier series of the function $f(x) = \begin{cases} 2, & -2 < x < 0 \\ x, & 0 < x < 2 \end{cases}$ converges to the value _____ at $x = 0$.	1	L1	2
2	If the Fourier expansion of the current $A = 4.4467 + (-4.0983 \cos \theta + 4.5767 \sin \theta) + (0.35 \cos 2\theta - 1.385 \sin 2\theta)$, compute the amplitude of first harmonic.	1	L3	1
3	Fourier series for the function $f(t)$ over the interval $(0, 5)$ is _____.	1	L1	2
4	Find the fundamental period of the function $f(x) = \cos x - \frac{1}{2} \cos 2x + \frac{1}{3} \cos 3x$.	1	L1	1
5	For the matrix $A = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix}$, find the Eigen values of the Matrix $A^T A$.	2	L1	2
6	If Fourier series of the function is $\sum_{n=1}^{\infty} \left\{ \frac{1}{2n\pi} [1 - (-1)^n] - \frac{4}{n^2\pi^2} \sin \frac{n\pi}{2} \right\} \sin n\pi x$, find the Fourier coefficients b_1 and b_2 .	2	L1	1
7	Draw the graph of the function $f(x) = \begin{cases} k+3, & -\pi < x < 0 \\ -(k+3), & 0 \leq x < \pi \end{cases}$, where $k > 0$. Hence find whether it is even or odd.	2	L2	2

PART - B

1	Apply singular value decomposition process to factorize the matrix $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$.	10	L3	2
2	An alternating current after passing through a rectifier has the form $I = \begin{cases} I_0 \sin x, & 0 \leq x \leq \pi \\ 0, & \pi \leq x \leq 2\pi \end{cases}$, where I_0 is the maximum current and the period is 2π . Express I as a Fourier series. Also draw the graph of the current periodically.	10	L2	4

MA231TA (Linear Algebra, Fourier Transforms and Statistics)

(CEE, EC, EI, ET)

PART-A

1. $\frac{1}{2} [2+0] = 1$

1M

2. $\sqrt{(-4.0983)^2 + (4.5767)^2} = 6.1435$

1M

3. $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{2n\pi x}{5} + b_n \sin \frac{2n\pi x}{5} \right)$

1M

$a_0 = \frac{2}{5} \int_0^5 f(x) dx, \quad a_n = \frac{2}{5} \int_0^5 f(x) \cos \frac{2n\pi x}{5} dx$

$b_n = \frac{2}{5} \int_0^5 f(x) \sin \frac{2n\pi x}{5} dx$

1M

4. $\angle CM \text{ of } (2\pi, \pi, 2\pi/3) = 2\pi$

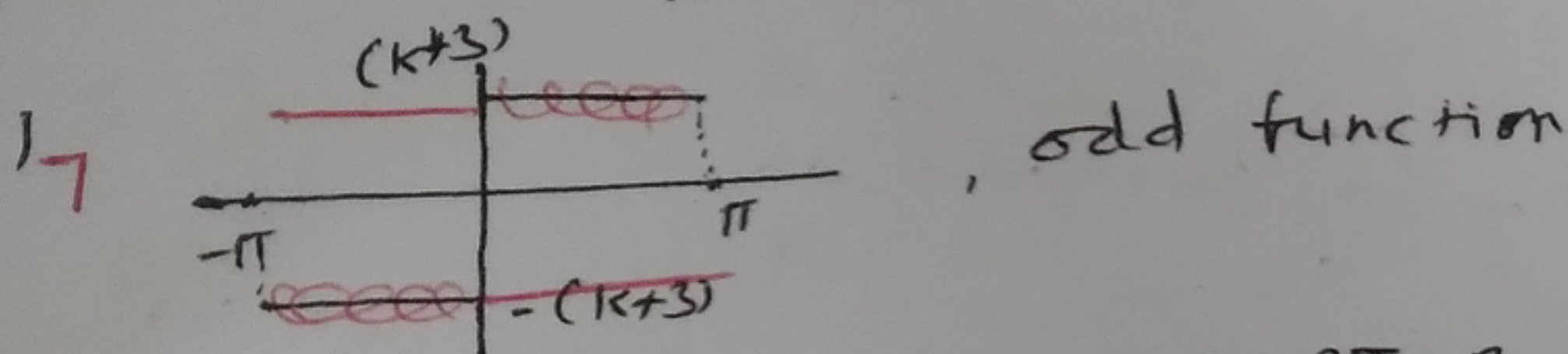
1+1M

5. $A^T A = [14] \Rightarrow |A^T A - \lambda I| = 0 \Rightarrow \lambda = 14$

1+1M

6. $b_1 = \frac{1}{\pi} - \frac{4}{\pi^2}, \quad b_2 = 0$

1+1M



PART-B

1. $A A^T = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad A^T A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix}$

1M

Eigen values of $A A^T$: $|A A^T - \lambda I| = 0, \lambda = 1, 3$

1M

Eigen vector for $\lambda = 1$: $\begin{bmatrix} -1 \\ 1 \end{bmatrix}$

2M

" " $\lambda = 3$, $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$U = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}, \quad \Sigma = \begin{bmatrix} \sqrt{3} & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$

2M

Eigen values of $A^T A$ are 3, 1, 0

Eigen vectors for 3, 1, 0 are $\begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$V = \begin{bmatrix} 1/\sqrt{6} & -1/\sqrt{2} & 1/\sqrt{3} \\ 1/\sqrt{6} & 1/\sqrt{2} & 1/\sqrt{3} \\ 2/\sqrt{6} & 0 & -1/\sqrt{3} \end{bmatrix}$

3M

$A = U \Sigma V^T$

1M

$$2) \quad I(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nx) + \sum_{n=1}^{\infty} b_n \sin(nx)$$

$$a_0 = \frac{1}{\pi} \int_0^{\pi} I_0 \sin x dx = \frac{2I_0}{\pi}$$

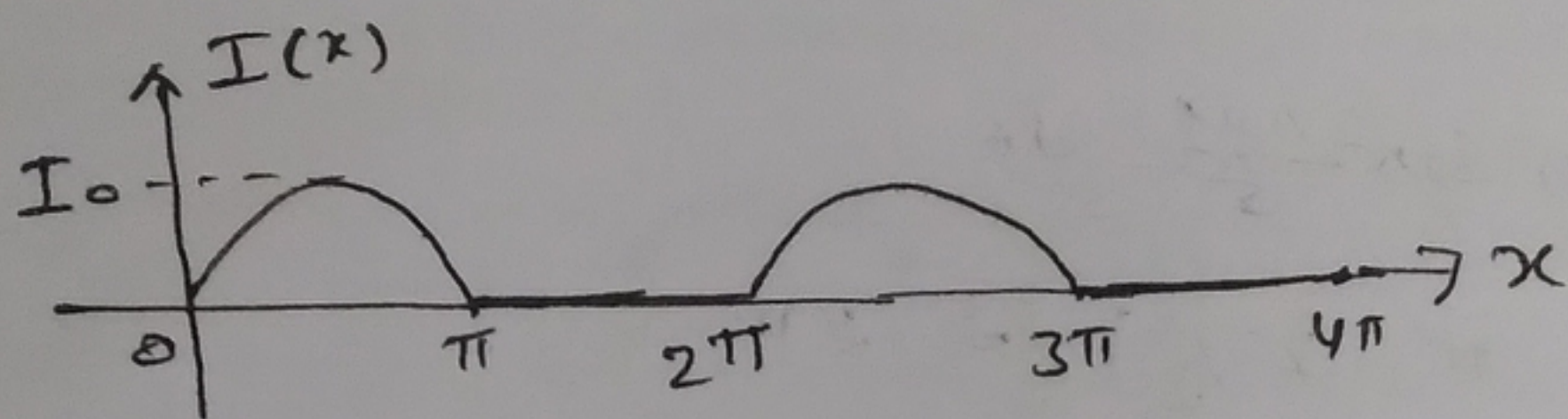
$$a_n = \frac{1}{\pi} \int_0^{\pi} I_0 \sin x \cos nx dx = \frac{-I_0}{\pi(n^2-1)} [(-1)^n + 1], \quad n \neq 1$$

$$a_1 = \frac{1}{\pi} \int_0^{\pi} I_0 \sin x \cos x dx = 0$$

$$b_n = \frac{1}{\pi} \int_0^{\pi} I_0 \sin x \sin nx dx = 0, \quad n \neq 1$$

$$b_1 = \frac{1}{\pi} \int_0^{\pi} I_0 \sin x \sin x dx = \frac{I_0}{2}$$

$$I(x) = \frac{I_0}{\pi} + \frac{I_0}{2} \sin x - \sum_{n=2}^{\infty} \frac{I_0}{\pi(n^2-1)} [(-1)^n + 1] \cos nx$$



3)

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{\frac{in\pi x}{2}}$$

$$c_n = \frac{1}{4} \int_{-2}^2 f(x) e^{-\frac{in\pi x}{2}} dx$$

$$= \frac{1}{4} \int_{-2}^0 (-a) e^{-\frac{in\pi x}{2}} dx + \int_0^2 (a) e^{-\frac{in\pi x}{2}} dx$$

$$= \frac{a}{in\pi} [1 - (-1)^n], \quad n \neq 0$$

$$c_{-n} = -\frac{a}{in\pi} [1 - (-1)^n], \quad n \neq 0$$

$$c_0 = \frac{1}{4} \int_{-2}^2 f(x) dx = \frac{1}{4} \left[\int_{-2}^0 (-a) dx + \int_0^2 a dx \right] = 0$$

$$f(x) = a + \sum_{\substack{n=-\infty \\ n \neq 0}}^{\infty} \left(\frac{-a}{n\pi} \right) [1 - (-1)^n] e^{\frac{in\pi x}{2}}$$

$$a_0 = 2c_0 = 0$$

$$a_n = (c_n + c_{-n}) = 0$$

$$b_n = i[c_n - c_{-n}] = \frac{2a}{n\pi} [1 - (-1)^n]$$

Trigonometric Fourier Series

$$f(x) = a + \sum_{n=1}^{\infty} \frac{2a}{n\pi} [1 - (-1)^n] \sin \frac{n\pi x}{2}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos n\pi x$$

$$= \frac{2}{1} \int_0^1 (1+x)^2 dx = \frac{2}{3} (8) = \frac{14}{3} \quad 2M$$

$$a_n = \frac{2}{1} \int_0^1 (1-x)^2 \cos n\pi x dx = \frac{4}{n^2 \pi^2} [2(-1)^n - 1] \quad 3M$$

$$f(x) = \frac{8}{3} + \sum_{n=1}^{\infty} \frac{4}{n^2 \pi^2} [2(-1)^n - 1] \cos n\pi x = (1+x)^2$$

$$\text{taking } x=0 \quad \sum \frac{1}{n^2} = \frac{\pi^2}{6} \quad 1M$$

$$4.6) f(x) = \sum_{n=1}^{\infty} b_n \sin(nx) \quad 1M$$

$$b_n = \frac{2}{\pi} \int_0^{\pi} e^{cx} \sin nx dx \quad 1M$$

$$= \frac{2}{\pi} \left[\frac{e^{cx}}{c^2 + n^2} \{ c \sin nx - n \cos nx \} \right]_0^{\pi} \quad 1M$$

$$= \frac{2n}{\pi (c^2 + n^2)} [1 - e^{c\pi} (-1)^n] \quad 1M$$

$$5) y = \frac{a_0}{2} + (a_1 \cos x + b_1 \sin x) + (a_2 \cos 2x + b_2 \sin 2x)$$

$$N = 6$$

$$\sum y = 8.7, \quad \sum y \cos x = -1.1, \quad \sum y \cos 2x = -0.3$$

$$\sum y \sin x = +0.5196, \quad \sum y \sin 2x = -0.1732 \quad 5M$$

$$a_0 = \frac{2 \sum y}{N} = 2.9$$

$$a_1 = \frac{2 \sum y \cos x}{N} = -0.37, \quad a_2 = \frac{2 \sum y \cos 2x}{N} = -0.1$$

$$b_1 = \frac{2 \sum y \sin x}{N} = 0.17, \quad b_2 = \frac{2 \sum y \sin 2x}{N} = -0.06 \quad 4M$$

$$y = 1.45 + (-0.37 \cos x + 0.17 \sin x) + (-0.1 \cos 2x - 0.06 \sin 2x)$$

$$y(x = \pi/4) = 1.2485 \quad 1M$$